Challenges Faced

- We faced challenges while loading the large SMS dataset from Hadoop HDFS into Apache Spark.
- Handling millions of SMS records required efficient data processing and memory management.
- Text preprocessing using Tokenizer and StopWordsRemover required proper configuration of the ML pipeline stages.
- Converting text messages into numerical features using HashingTF and IDF was an important implementation challenge.
- We encountered Python serialization errors while working with PySpark and debugging the pipeline execution.
- Saving and loading the trained Spark ML Pipeline model required correct model-handling methods.
- We resolved these issues through debugging and configuration changes and successfully trained and evaluated the SMS spam detection model.